

Computer Graphics

CSIT — 3rd Semester — 2079 — Answer Sheet
Subject Code: CSC214

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain the working details of Mid-Point Circle Algorithm? Trace the points along the line path having two end points (6, 9) and (2,3) using Bresenham's line drawing algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Differentiate between object space and image space method of hidden surface removal. Describe the Z-buffer hidden surface removal algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Write the algorithm for Cohen-Sutherland Line clipping. Clip the polygon A (100,150), B (200,250) and C (300,200) with clipping window defined by the coordinates (100,300), (300,300) and (200,100) using Sutherland Hodgeman Polygon Clipping Algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Reflect a line segment having end points (9,3) and (12,10) about a line $Y = 7$. Draw initial and final result graph as well.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Differentiate between Raster and Vector graphics method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain about parametric curve. Describe the properties of Bezier curve.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are blobby objects? How it is represented? Explain the wireframe representation of 3D objects.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Calculate the total memory required to store a 8 minute video in a SVGA system with 24 bit true color and 25 fps.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Find the composite transformation matrix for reflection about a line $y = mx + c$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is polygon table? Explain the use of this method in 3D object representation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define the term "Rendering" in computer Graphics. Explain Phong Shading Method with its advantage and disadvantage.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on (Any TWO)a. BSP Treeb. Virtual Realityc. Intensity Attenuation

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.